

Exchange Rate Volatility and Export Trade in Nigeria: An Empirical Investigation

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Abstract

The paper seeks to quantitatively assess the impact of exchange rate volatility on non oil export flows in Nigeria. Theoretically, volatility-trade link is ambiguous, although a strand of studies reported inverse link between export flow and volatility. The paper employed fundamental analysis where the flow of non oil exports from the Nigerian economy is assumed to be predicated on fundamental variables: the naira exchange rate volatility, the US dollar volatility, Nigeria's terms of trade (TOT) and index of openness (OPN). Empirical results showed presence of unit root at level, however, the null hypothesis of nonstationarity was rejected at first difference. Cointegration results revealed that a stable long run equilibrium relationship exists between non oil exports and the fundamental variables. Using quarterly observations for twenty years, vector cointegration estimate revealed that the naira exchange rate volatility decreased non oil exports by 3.65% while the same estimate for the US dollar volatility increased export of non oil in Nigeria by 5.2% in the year 2003. The paper recommends measures that would promote greater openness of the economy and exchange rate stability in the economy.

Keywords: exchange rate volatility, non oil exports, terms of trade, index of openness, unit root and cointegration analysis.

1.0 Introduction

Research related to exchange rate management still remains of interest to economists, especially in developing countries, despite a relatively enormous body of literature in the area. This is largely because the exchange rate in whatever conceptualization, is not only an important relative price, which connects domestic and world markets for goods and assets, but it also signals the competitiveness of a country's exchange power vis-à-vis the rest of the world in a pure market. Besides, it also serves as an anchor which supports sustainable internal and external macroeconomic balances over the medium-to-long term. There is, however, no simple answer to what determine the equilibrium exchange rate, and estimating equilibrium exchange rates and the degree of exchange rate misalignment remains one of the most challenging empirical problems in open-economy macroeconomics (Williamson, 1994).

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The fundamental difficulty is that the equilibrium value of the exchange rate is not observable. While the exchange rate misalignment refers to a situation in which a country's actual exchange rate deviates from such an unobservable equilibrium, an exchange rate is said to be "undervalued" when it depreciates more than its equilibrium, and "overvalued" when it appreciates more than its equilibrium. The issue is, unless the "equilibrium" is explicitly specified, the concept of exchange rate misalignment remains subjective. The problem of subjectivity is, especially so, according to Chang and David (2005) because exchange rate equilibrium or misalignment is measured over different time horizons. Notwithstanding, Edwards (1989) states that the equilibrium real exchange rate (*RER*) prevails when given sustainable values for other relevant variables, such as terms of trade, capital and aid flows, and technology, the economy achieves both *internal* and *external* equilibrium.

There is growing agreement in the literature that prolonged and substantial exchange rate misalignment can create severe macroeconomic disequilibria and the correction of external balance will require both exchange rate devaluation and demand management policies. The main intuition behind this is that an increase in exchange rate volatility leads to uncertainty which might have a negative impact on trade flows or according to Anderton and Skudely (2001) the economic logic underpinning the negative link is the aversion of firms to engage in a risky activity, namely trade. Baldwin, Skudelny and Taglioni (2005) discovered that the effect of exchange rate uncertainty on trade in the European Union (EU) countries is negative; trade increases as volatility falls and gets progressively larger as volatility approaches zero. While numerous studies were conducted on the extent of naira exchange rate and its misalignment in Nigeria (see Soludo and Adenikinju, 1997; Agu, 2002; Omotosho and Wambai, 2005; Obaseki, 2001; CBN, 2007a; CBN, 2007b; CBN, 2008), assessment of the impact of exchange rate volatility on export has in the recent past been nonexistent.

Against this background, this paper seeks to quantitatively measure the impact of exchange rate volatility on non oil export trade in Nigeria from 1986Q1 to 2006Q4. The rest of the paper is organized as follows. Section two presents a survey of the literature and theoretical issues relating to exchange rate volatility and trade flows. Section three discusses the

methodology employed in the study while section four analyses the empirical results. Finally, section five contains conclusions and recommendations.

2.0 Literature Review and Theoretical Issues

The traditionalist view on the impact of currency depreciation on trade indicates that it leads to an expansion in trade via lower export prices. The structuralist school, however, stresses some contractionary effects, Meade (1951). Hirschman (1949) points out that currency depreciation from an initial trade deficit reduces real national income and may lead to a fall in aggregate demand. Kandil and Mirzaie (2002) argued that currency depreciation gives with one hand, by lowering export prices and takes away with the other hand, by raising import prices. They observed that if trade is in balance and terms of trade remain unchanged, these price changes offset each other, especially when the famous Marshall-Lerner² condition is not satisfied. If imports exceed exports, the end result is a reduction in real income within a country, Cooper (1971). See Diaz-Alejandro (1984), Krugman and Taylor (1978) and Edward (1986)

Recently, it is a widely accepted tenet that chronic misalignment in the real exchange rate has been a major source of slow growth in Africa and Latin America, while prudent macroeconomic, trade and exchange rate policies have fostered growth in Asia (World Bank, 1984; Edwards, 1988; Ghura and Grennes, 1993; Rodrik, 1994 and Yotopoulos 1996). According to Yotopoulos and Sawada (2005), systematic deviations of nominal exchange rate (NER) from their purchasing power parity (PPP) levels may engender serious instabilities of the international macroeconomic system. According to Baldwin, Skudelny and Taglioni (2005), disequilibrium exchange rate values have been conclusively shown to have negative link with trade (see inter alia, European Commission, 1995). Some authors, however, argue that under the existence of forward exchange markets, exchange rate uncertainty can be completely covered so that there is no impact of exchange rate uncertainty on trade (Ethier, 1973 and Baron, 1976). However, Viaene and de Vries (1992) argued that even under the forward exchange markets there may be an indirect effect of exchange rate volatility on trade if hedging is costly.

² A condition when exchange rate changes restore equilibrium in BOP by devaluing a country's currency. This holds when the sum of price elasticities of demand for exports and imports in absolute terms is greater than unity, devaluation will improve the country's BOP, that is, $e_x + e_m > 1$.

Empirical studies in the past that applied time series analysis and found no significant relationship between volatility and trade. The few that found a link suggest that the effect was very small (see Khan (1974), Koray and Lastrapes (1989), Belanger and Gutierrez (1998), Bini-Smaghi (1991), Kenen and Rodrik (1986) and Sekkat (1998). Meese and Rogoff (1983), in a work which predates the cointegration literature, forecast exchange rates by simply regressing the exchange rate on the macroeconomic fundamentals and then using these parameter estimates and the ex post realized and revised values of the future economic fundamentals to predict the future exchange rate. Cross-sectional studies were also carried out by Hooper and Kohlhagen (1978), De Grauwe (1987), Brada and Méndez (1988), De Grauwe and Verfaillie (1988), Savvides (1992), Sapir, Sekkat and Weber (1994) and Eichengreen and Irwin (1995) find evidence of a negative effect of exchange rate uncertainty on export. Again, this effect, in most cases, was relatively small.

Some studies employed co-integration analysis, for example, Koray and Lastrapes (1989), Arize (1997, 1998a and b), Fountas and Aristotelous (1999) and Flam and Jansson (2000). A detailed empirical review of this strand of literature is reported in Baldwin, Skudsen and Taglioni (2005). The results of the studies taking into consideration the trend characteristics of the time-series appeared to be more clear-cut and most suggest a significant negative effect of exchange rate uncertainty on the trade variables. For instance, Fountas and Aristotelous (1999) found a significant negative long run effect of exchange rate uncertainty on trade. Wei (1999) found a negative and statistically significant effect for foreign exchange rate volatility on exports taking account of futures and options instruments to hedge risk. Recently, Baum *et al* (2004) showed evidence of a positive relationship between exchange rate volatility and trade using a *poisson* flexible lag structure, while Klaassen (2004) did not find evidence of any significant effect of exchange rate volatility on trade for G7 economies.

Caporale and Doroodian (1994) used a generalized autoregressive conditional heteroskedasticity (GARCH) technique to measure the volatility of exchange rate and discovered significant negative effect of volatility on import trade. McKenzie and Brooks (1997) and McKenzie (1999) used ARCH modeling and introduced an exchange rate volatility term into their export trade models for both German-US and Australian trade flows respectively. Their results were statistically significant but, showed positive impact of volatility on trade, while for McKenzie (1999), the results were mixed.

Furthermore, studies that employed panel estimation techniques, according to Anderton and Skudelny (2001) emerge with better results. For example, Abrams (1980), Thursby and Thursby (1987), Dell’Ariccia (1998), Pugh, *et al* (1999) and Rose (1999), all found significant negative effect of the proxy for exchange rate uncertainty. In particular, while Dell’Ariccia (1998) found that the trade gains resulting from the elimination of exchange rate volatility would have been 10 percent. Anderson and Skudelny (2005) discovered that exchange rate volatility would decrease extra-euro area imports by around 10 percent.

Another strand of empirical studies apply gravity-type trade model to assess the impact of exchange rate volatility on bilateral trade. Pugh, *et al* (1999) use 16 OECD countries and showed that volatility leads to a once and for all decrease in the level of trade by around 8 percent and Rose (2000) estimated a gravity trade model for 186 countries using a 5-year moving average of the variance of the nominal exchange rate return and discovers that exchange rate volatility has a significant negative impact on trade (estimates show that zero exchange rate volatility would have resulted in a 13 percent increase in trade). It was this seminal work (Rose (2000)) that started the debate that countries participating in a currency union seemed to trade three times more than expected – even when one controls for the impact of exchange rate volatility. This discovery was christened the *Rose effect*. Rose and Engel (2002) and Glick and Rose (2002) found empirical evidence in support of the *Rose effect*. Furthermore, Aliyu (2007a) uses a gravitational model for Nigeria-India bilateral trade and discovered that the exchange rate coefficient is theoretically consistent and statistically significant in the import model for the Indian economy but not for the Nigerian economy.

A number of empirical studies on Nigeria were carried out by Ojo, *et al* (1978), Osagie (1985), and all downplayed the role of exchange rate in the import-export trade in the country. This was largely possible in view of the system of exchange rate regime prior to the introduction of structural adjustment programme in Nigeria in July 1986. However, Oyejide (1986), Omolola (1992), Akanji (1992), Ihimodu (1993) Osuntogun, *et al* (1993) World Bank (1994), Aliyu (1994 & 2001) discovered that exchange rate depreciation caused significant changes in the structure and volume of Nigeria’s agricultural exports. Egwaikhede (1999) in his dynamic specification model of import determinants in Nigeria from 1953 to 1989 discovered that short run changes in the availability of foreign exchange earnings, relative prices, and real output (income), significantly explained the growth in total imports in Nigeria. On exchange rate instability, Nnanna (2002) links exchange rate instability in

Nigeria to adverse monetary policy outcome, inflation, interest rate and growth in money supply; and the failure of monetary policy was linked to fiscal dominance in the economy. Aliyu (2007b) showed that exchange rate significantly affects imports more than exports due largely to the monocultural nature of Nigeria's exports and inexhaustible and multifarious nature of its imports. According to a study by the CBN (2007) using fundamental variables; TOT, nominal effective exchange rate (NEER) and lagged real exchange rate; findings suggest that the three variables accounted for 22, 55 and 99 percent of variations in the dependent variable, respectively.

Theoretically, the volatility-trade link is ambiguous according to Baldwin, Skudelny and Taglioni (2005). Dornbusch (1993) observed that the effect of an appreciated exchange rate on trade would be to make production of tradable unprofitable and non-tradable goods more profitable. In other words, imports will be high, while exports will tend to be discouraged. Cottani, *et al* (1990) found that misalignment was strongly related to lower per capita GDP growth, and to low productivity, slow export growth and slow agricultural growth. Loaza, *et al* (2002) also found a negative relationship between overvaluation and growth, holding other macroeconomic variables constant³.

It is evident from the above review that studies on the impact of exchange rate volatility on trade have no dominant approach. The choice of a particular approach or methodology and expected outcomes depend on a particular economy and nature and availability of data. Gala and Luccinda (2006) state that two main methods of dealing with exchange rate misalignment are the purchasing power parity (PPP) approach and fundamental analysis. The PPP approach, on one hand, is based on relative prices and considers high international price levels as proxy for exchange rate overvaluation for a given GDP per capita level. Fundamental analysis, on the other hand, considers economic fundamentals in modeling exchange rate misalignment. These include terms of trade (TOT), balance of payments (BOP) financing condition, fiscal policy stance (surplus or deficit spending), degree of openness (OPN), GDP per capita, etc.

³ For more extended review of literature on the effect of exchange rate volatility on trade, see IMF (1984), Cote (1994), McKenzie (1999), Shatz and Tarr (2001), Skudelny (2002) and Taglioni (2002).

It has also been established in the literature that a drop in exchange rate volatility can increase the volume of trade in two not mutually exclusive ways – by producing more exports, and by increasing the number of firms that are engaged in exporting. It is this theorization that accounts for a negative volatility-trade link, Baldwin, Skudelny and Taglioni (2005). Generally, the transmission mechanism through which exchange rate volatility affects non oil exports in Nigeria could be both from the supply and demand channels. The supply side effects are related to the fact that exchange rate volatility could affect input prices. This induces some producers to lower output and in the face of volatile exchange rate, makes the exports less competitive. Exchange rate volatility could also affect consumer confidence in importing countries and thus lowers demand. It also adversely affects investment indirectly by increasing producers' cost. Against this background, this paper seeks to assess the link between exchange rate and non oil export trade performance in Nigeria. Other additional variables would too incorporated in the model.

3.0 Research Methodology

In line with the methodology employed by Koray and Lastrapes (1989), Arize (1997) and (1998a and b) and Fountas and Aristotelous (1999), this paper adopts a vector error correction (VEC) methodology in analyzing the effect of exchange rate volatility on Nigeria's non oil exports between 1986Q1 and 2006Q4. Total non oil exports (*nexp*) in Nigeria is assumed to follow the path dictated by fundamentals such as exchange rate volatility in Nigeria (*vol_n*) and Nigeria's trading partner (*vol_p*) (the United States' dollar volatility was used as proxy), Nigeria's terms of trade (*tot*) and index of openness (*opn*). The paper uses the Johansen's cointegration analysis to identify the long run relationships among the variables. Before estimating the cointegrated VAR by Johansen's method, the stochastic properties of the data was checked using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests. In the ADF test, the hypothesis $\delta = 0$ or $\rho = 1$ of nonstationarity or unit root is tested against the alternative which states that a series is stationary if $-1 < \rho < 1$. The PP test on the other hand uses nonparametric statistical methods to account for the serial correlation in the error term, without necessarily adding lagged difference terms as in the ADF case. Appropriate lags were selected on the basis of Schwarz information criteria (SIC) in order to ensure independence in the residual series.

The purpose of the cointegration test is to determine whether a group of nonstationary series is cointegrated or not and as a starting point, the presence of a cointegrating relation forms the basis of the VEC specification. Johansen (1991, 1995) developed a VAR-based cointegration tests the specification of which runs as follows:

Consider a VAR of order:

$$y_t = A_1 y_{t-1} + \dots + A_p y_{t-p} + Bx_t + \varepsilon_t \quad (1)$$

Equation (1) is saying that y_t is a k vector of endogenous variable, x_t is a d vector of exogenous variables (constants, trends and dummies), $A_1 \dots A_p$ and B are matrices of coefficients to be estimated, and ε_t is a vector of innovations or impulses or shocks. Rewriting the above equation in the following VEC form:

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + Bx_t + \varepsilon_t \quad (2)$$

Granger's representation theorem asserts that if the coefficient matrix Π has reduced rank $r < k$, then there exist $k \times r$ matrices α and β each with rank r such that $\Pi = \alpha \beta'$ and $\beta' y_t$ is $I(0)$. r is the number of cointegrating relations (*the cointegrating rank*) and each column of β is the cointegrating vector. The elements of α are known as the adjustment parameters in the VEC model. Johansen's method is to estimate the Π matrix from an unrestricted VAR and to test whether we can reject the restrictions implied by the reduced rank of Π .

Meanwhile, the first step is to establish the order of integration of each of the variables. It is only then that they would form the basis for estimating the long run relationship between them and if not, to establish the order of their integration or the number of times they have to be differenced to become stationary. For the purpose of this study, the variables used in the analysis are: *nexp*, *vol_n*, *vol_p*, *tot* and *opn*. The cointegration equation, with all the series converted into natural log, is expressed as follows:

$$\ln exp = \alpha_0 + \beta_1 \ln vol_n + \beta_2 \ln vol_p + \beta_3 \ln tot + \beta_4 \ln opn + \varepsilon_t \quad (3)$$

By converting them into log, we are interested in measuring the rate of change, which would be captured by the coefficients of the regressors. Equation (3) is thus saying that total non oil exports in Nigeria is explained by the right hand side variables, which were earlier defined.

The equation is estimated by system of least squares. The next section is on definition of variables.

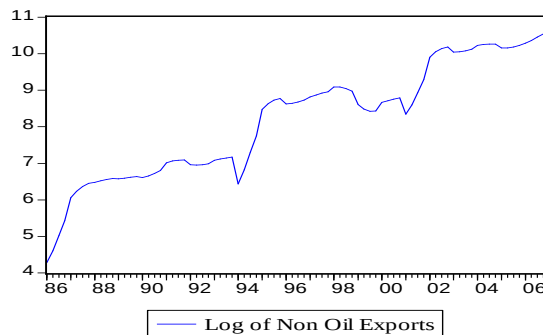
3.1 Measurement of Variables Used in the Estimation

As stated earlier in the theoretical foundation of the model, the fundamental variables that were used to explain the pattern of Nigeria's non oil exports trade include naira and US dollar exchange rate volatility, Nigeria's term of trade and index of openness. The variables are defined as they were applied in the analysis.

Total Non Oil Exports

Data on Nigeria's total non oil exports (fob) were obtained from various issues of Central Bank of Nigeria (CBN) Statistical bulletin in nominal terms from 1986Q1 to 2006Q4. This was converted into natural log and was tested for stationarity. The data was tested for unit root and was then differenced d times to attain stationarity. Figure 1 presents the graph of the log of quarterly series of total non oil exports in Nigeria from 1986Q1 to 2006Q4.

Figure 1



Exchange Rate Volatility

There is armful literature on the measurement of exchange rate volatility. Depending on the approach one adopts, Anderton and Skudelny (2001), for instance, measured exchange rate volatility as the quarterly variance of the weekly nominal exchange rate while Zubair and Jega (2008) measured volatility by the standard deviation of each series through their sample. Gujarati (2003) suggested the use of mean-adjusted and the squared deviation of (variance) of each series in a sample. This paper, in line with Zubair and Jega's paper measures exchange

rate volatility as the standard deviation of each series of quarterly observation from the average nominal exchange rate of the naira vis-à-vis the US dollar.

$$voln = \sqrt{\Sigma(NER_{ij} - \overline{NER}_j)^2} \quad (4)$$

Quarterly data on nominal exchange rate of the naira was obtained from the Research Department of the CBN. This measurement approach allows the VEC model to capture not only current volatility but, contemporaneously along with some history of past volatility when the model is opened to higher lag orders. The standard deviation series was converted in to natural log and then tested for stationarity. The same procedure was applied to arrive at the measure of volatility of the US dollar. The nominal effective exchange rate (NEER) in the case of the US was obtained from the IFS (nec) on quarterly basis and the standard deviation was computed there from and was tested for stationarity as well. Theoretically, volatility relates inversely with real export, but, empirical findings suggest that the coefficient could also bear positive sign. Figures 2 and 3 present the graph of the Nigeria's naira and the US dollar volatility

Figure 2

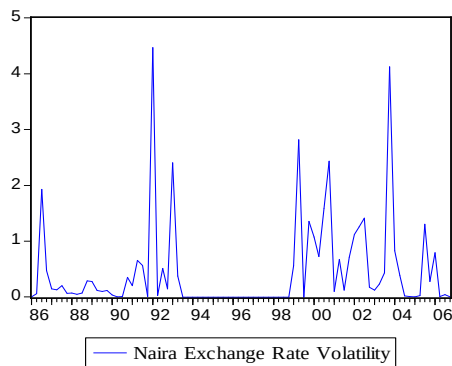
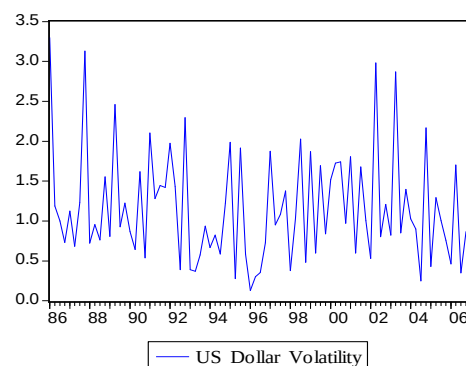


Figure 3



Terms of Trade

This is computed as the ratio of export price index (P_x) to import price index (P_m). Baffes, Elbadawi and O'Connell (1997) develop a measure of terms of trade and trade policy. The measure in addition to tot , captures the domestic trade policy stance. It is given as:

$$\frac{P_x}{P_m} = \frac{\varphi}{\eta}, \quad \varphi \equiv \frac{P_x}{P_m}, \quad \eta \equiv \frac{1 + t_m}{1 + t_x} \quad (5)$$

Where t_m and t_x measure tariff on imports and exports respectively. For the purpose of this analysis, only the ratio of the Nigeria's consumer price index and the US's producer price index is taken as proxy as former's terms of trade. The two series were obtained from CBN's statistical bulletin and international financial statistics (IFS) line 63a for Nigeria and US respectively. The base year of the two series was adjusted to 2000 = 100. The series was tested for unit root and has to be differenced to attain stationarity. It is expressed as the differenced form of log of terms of trade (*dltot*).

Figure 4

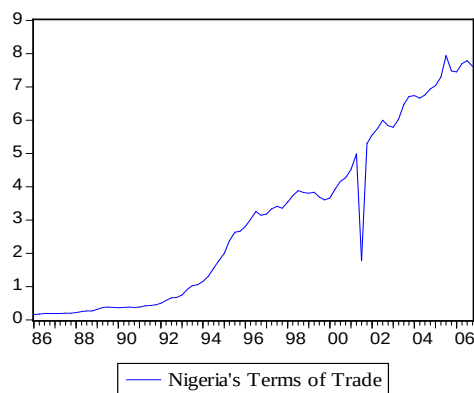
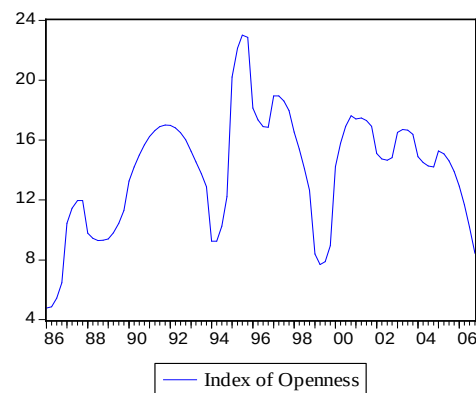


Figure 5



Note that the measure did not, however, incorporate such information on the level of import and export taxes largely due to the difficulty in obtaining information on them and /or reliable information on their levels. The above figure 4 presents the graph of the log of terms of trade.

Index of Openness

The variable is measured as the sum of total trade, imports and exports divided by gross domestic product. Data was obtained from various issues of Statistical Bulletin published by the Central Bank of Nigeria. The series was converted into log and was tested for stationarity. Theoretically, an increase in openness is assumed to be arising from a decline in tariff rates, leading to a fall in the domestic prices of importables. This will lead to high demand of foreign currency (to take advantage of cheap imports), and less demand for domestic currency. Hence this is expected to lessen exchange rate volatility, increase competitiveness and promote more exports. As a result, the openness variable is expected to carry a positive sign. Figure 5 presents the graph of the log of index of openness in Nigeria.

4.0 Results and Discussions

From the outset, this section starts with the presentation of results of unit root tests. This was followed with the cointegration test based on the specification given in equation (2). Table 1 summarizes the results of the ADF and the PP tests applied to the variables. When tested at levels, the volatility measures - *voln* and *volp*, were found to be stationary while the others were not. At first difference, however, the others, that is, *lnexp*, *ltot* and *lopn* were stationary at 1 percent for the first two and 10 percent level for the latter using ADF test with a constant.

Table 1: Augmented Dickey Fuller and Phillips-Perron Stationarity Tests

Variable/ coefficient	ADF- Test				Phillips- Perron Test			
	With Intercept		With intercept & Trend		With Intercept		With intercept and Trend	
	t-Statistic	Decision Rule	t-Statistic	Decision Rule	t-Statistic	Decision Rule	t-Statistic	Decision Rule
<i>lnexp</i>	-4.20*	I(1)	-4.13*	I(1)	-5.94*	I(1)	-5.96*	I(1)
<i>lvoln</i>	-8.24*	I(0)	-8.31*	I(0)	-8.27*	I(0)	8.31*	I(0)
<i>lvolp</i>	-12.14*	I(0)	-12.07*	I(0)	-11.94*	I(0)	-11.88*	I(0)
<i>ltot</i>	-10.04*	I(1)	-10.21*	I(1)	-17.84*	I(1)	-21.19*	I(1)
-								
<i>lopn</i>	2.61***	I(1)	2.97	N.E	-5.33	I(1)	-5.45	I(1)

Note: One and three asterisks denote rejection of the Null hypothesis at 1%, and 10%, respectively, based on the MacKinnon critical values.

Thus, the null hypothesis of nonstationarity or unit root is rejected. The next step is to test for cointegration or the long run relationship between the real export and its fundamentals.

4.2 Cointegration Results

The result of the unrestricted Johansen cointegration test using the specification in equation (2) is presented in Table 2. The standard statistics used in the interpretation of the test are the eigenvalue and the trace statistic at given level of significance.

Results in Table 2 show the existence of only one cointegration equation on the basis of trace statistic. The presence of one cointegration unveils the existence of a long run equilibrium relationship between real non oil export and the fundamentals used in the model. The

hypothesis of no cointegration could not, however, be rejected on the basis of maximum eigenvalue because the hypothesized value is greater than the calculated.

Table 2: Johansen Cointegration Test

Maximum Rank/ Number of Cointegrating Equations	Maximum Eigenvalue	Critical Value (Eigenvalue)	Trace Statistic	Critical Value (Trace Statistic)	Probability**
0*	31.64	33.88	79.49	69.82	0.0069
1	21.68	27.58	47.86	47.86	0.0500
2	17.15	21.13	26.18	29.80	0.1236
3	9.021	14.26	9.025	15.49	0.3631
4	0.004	03.84	0.004	03.84	0.9465

Trace test indicates 1 cointegrating equations at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Although not encountered here, but, the existence of multiple cointegrating vectors complicates the interpretation of an equilibrium condition (Johansen and Juselius, 1992, Dibooglu and Enders, 1995, Wickens, 1996, MacDonald and Nagayasu, 1998, Clark and MacDonald, 1999). Neither is the case of a single cointegrating vector the most desired outcome because such makes it unclear if the vector represents a structural or reduced form relationship. Therefore, while interpreting the cointegrating vectors obtained from the Johansen procedure as was pointed out by Cheng and Orden, 2005 and Ilimi, 2006, one needs to note that what the reduced rank regression provides is information on how many unique cointegrating vectors *span* the cointegrating space, while any linear combination of the stationary vectors is itself a stationary vector.

Thus, from the above, we apply the Johansen procedure to obtain the long run coefficients of the model. Table 3 presents the normalized coefficients (β) of the variables in the model. All the coefficients were correctly signed and statistically significant at 1 the percent level. The first two coefficients of the naira and US dollar volatility have negative and positive signs respectively. This implies that while naira volatility adversely affects non oil exports, volatility in the US dollar promotes it. These findings are consistent with those reported by Baum *et al* (2004) who discovered positive link between export and volatility on one hand and those reported by Caporale and Doroodian (1994), Pugh, *et al.* (1999), Wei (1999), Rose and Engle (2002), Anderton and Skudelny (2005) on the other hand. However, McKenzie and

Brooks (1997) and McKenzie (1999) using ARCH modeling reported mixed effects of exchange rate volatility on the level of exports.

Table 3: Normalized cointegrating Eigenvector (β')

<i>Unrestricted</i> <i>Coefficients</i>	
Cointegrating Equation:	CointEq1
<i>lnexp(-1)</i>	1.000
<i>lvoln(-1)</i>	-0.885* (-4.87)
<i>lvolp(-1)</i>	1.819* (4.31)
<i>ltot(-1)</i>	0.509* (14.4)
<i>lopn(-1)</i>	0.091* (3.23)
<i>C</i>	3.726

() report values of t- ratios

* Indicates significance at 1% level

The result further shows a strong and statistically significant positive link between exports and both the terms of trade and index of openness are export enhancing. Based on the estimated cointegrating vector β , the long-run equilibrium equation can be written as:

$$lnexp = 3.726 - 0.885*lvoln + 1.819*lvolp + 0.509*ltot + 0.091*lopn \quad (6)$$

(0.205) (0.373) (0.035) (0.028)

The above cointegrating equation reveals a negative relationship between non oil exports and the naira exchange rate volatility while positive relationship subsists in the others. Economically speaking, a volatile currency could hamper inter temporal contracts and could have both supply and demand implications. A weak naira, for instance, could, all things being equal, make Nigeria's non oil exports highly competitive⁴, although at the same time, this could have serious supply side implications by way of increase in the cost of production at industry and firm levels. Equally, the sign of coefficient of dollar volatility along with those

⁴ This is, however, not to discount the famous argument of Singer-Presbisch thesis, which unrevealed deteriorating conditions in the terms of trade in developing countries in their trade with the developed nations. Although the thesis has lost some of its relevance in the last 30 years, when developing countries outside of Africa begin exporting simple manufacture, Nigeria's non oil exports still compose large components primary exports.

of terms of trade and openness suggest that non oil exports in Nigeria is positively affected in the long run by a change in any of the regressors.

The above findings are reinforced by the results reported in Table 4 showing alpha adjustment coefficients of the model. Here the alpha is viewed as the speed of adjustment parameter. Meaning, if the system is out of long run equilibrium condition, adjustment comes through the alpha. Therefore, the numerical value and statistical significance of each α coefficient is very important in the evaluation of the extent of speed of adjustment for any shock that destabilizes the long run equilibrium condition.

Table 4: Alpha Adjustment Coefficients
(Standard errors in Parenthesis)

Variable	Coefficient & Standard Errors
<i>lnexp(-1)</i>	-0.054 (-1.05)
<i>lvln(-1)</i>	-0.731* (2.79)
<i>lvlp(-1)</i>	0.532* (2.80)
<i>ltot(-1)</i>	0.205 (1.64)
<i>lopn(-1)</i>	-0.212 (-0.60)

* denote significance at 1%

Based on restriction imposed on *lnexp* alpha coefficient, the result shows that the two volatility adjustment parameters were statistically significant. The results imply that if the system is out of equilibrium condition, adjustment back to steady state comes from the two volatility coefficient – naira exchange rate and dollar volatility. The former decreases while the latter increases in the restoration of the equilibrium condition. This attests to the influence of exchange rate volatility on the level of non oil export in Nigeria in the long run. To assess the short run equilibrium dynamics, a vector error correction model was estimated by incorporating an error correcting mechanism in the cointegrating equation (3). The error term was obtained from a conventional regression using ordinary least squares (OLS) method applied to the same equation. The results presented in Table 5 show that the error correction variable is correctly signed and significant at 1 percent level.

Table 5: Short run Vector Error Correction Model

Dependent Variable: DLNEXP				
Method: Least Squares				
Date: 02/16/09 Time: 09:38				
Sample (adjusted): 1986Q3 2006Q3				
Included observations: 81 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
<i>ecm(-1)</i>	-0.20186*	0.058652	-3.44166	0.0010
<i>lvoln</i>	-0.01748	0.022357	-0.78181	0.4369
<i>lvolp</i>	-0.01950	0.031484	-0.61949	0.5376
<i>dltot</i>	0.10893*	0.039441	2.76196	0.0073
<i>dlopn</i>	0.08328*	0.012453	6.68777	0.0000
<i>lvoln(1)</i>	-0.00182	0.022662	-0.08019	0.9363
<i>lvolp(-1)</i>	-0.00136	0.032086	-0.04251	0.9662
<i>dltot(-1)</i>	0.07362	0.038036	1.935696	0.0569
<i>dlopn(-1)</i>	-0.02112	0.012864	-1.64214	0.1050
<i>C</i>	0.08636	0.058976	1.464429	0.1475
R-squared	0.469063	Akaike info criterion		-0.71953
Adj. R-squared	0.401762	Schwarz criterion		-0.42391
SSR	1.804229	F-statistic		6.96954
D.W	1.198330	Prob. (F-statistic)		0.00000

* indicates significance at 1 percent level.

The results confirm that non oil exports in Nigeria has an automatic adjustment mechanism and that non oil exports in Nigeria responds to deviations from equilibrium in a balancing manner. A value of -0.20186 for the *ecm(-1)* coefficient suggests that a fast speed of adjustment of roughly eight quarters or two years⁵. See appendix for normality and residual tests.

4.3 Impact of Exchange Rate Volatility on Non oil Exports

The above analysis, beside the fact that the coefficients of volatility measure sensitivity of non oil exports to a shock in exchange rate; we can also use the same coefficients to assess the impact of volatility on the level of non oil exports in real terms. The paper, calculates the impact of exchange rate volatility by multiplying the values of exchange rate volatility variables over the sample by the respective vector coefficients of the naira and US dollar

⁵ The coefficients measure the average number of times that a given shock is corrected in the model. This is given as $(1 - \alpha)^t$, which is, $(1 - \alpha)$, where t is the number of years and α is the absolute value of the adjustment parameter.

volatility, that is, -0.885 and 1.819 respectively. Figure 5 shows that the naira exchange rate volatility reduced non oil exports by about 4 percent in the first quarter of 1992 and by 3.65 percent in 2003Q4. The impact was nil between 1993Q4 and 1998Q4 because an exchange rate of N21.886 to a US dollar was maintained throughout the range. The average impact for the entire sample is 0.45 percent

Figure 5: Impact of Naira Volatility on Non oil Exports

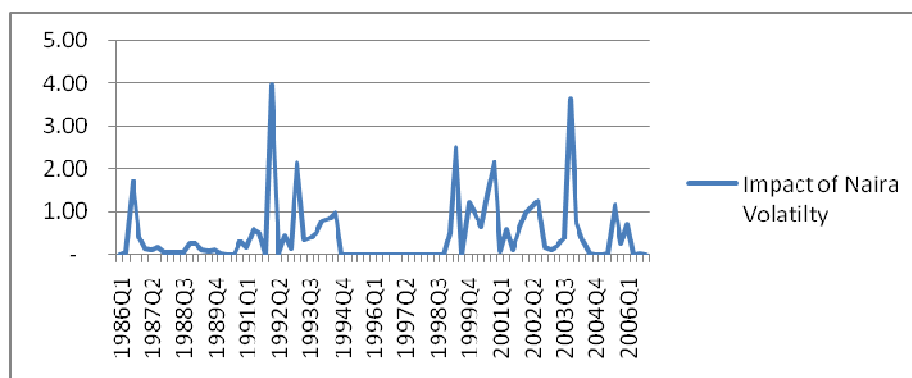
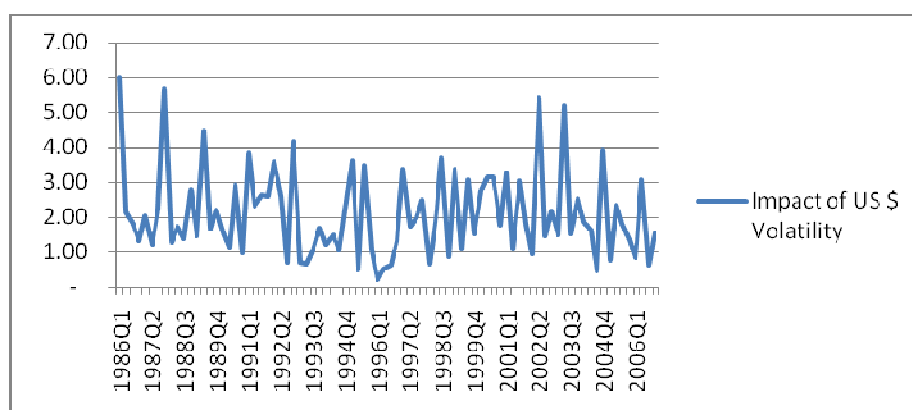


Figure 6 shows that the impact of dollar volatility on non oil exports in Nigeria is a bit more violent relative to that of the naira volatility. The figure shows that the average impact stood at 2.1 percent between 1986Q1 to 2006Q4. Specifically, evidence shows that the US dollar volatility may have increased non oil exports in Nigeria in 1986Q1 by up to 6 percent. Others include by 5.4 percent in 2002Q2 and 5.2 percent 2003Q2. Similar findings were reported elsewhere by Anderton and Skudelny (2001) in their analysis of trade effect of the euro. For instance, they showed using the same approach that extra-euro exchange rate volatility may have decreased extra-euro imports by 10 percent.

Figure 6: Impact of Dollar Volatility on Non Oil Exports



Above findings are valid and consistent with some early empirical studies on assessment of impact of structural adjustment programme (SAP) on non oil sector export subsector in Nigeria, Omolola (1992), Akanji (1994) and Aliyu (1994). The studies discovered that persistent exchange rate depreciation – above evidence suggests that naira is volatile because it persistently loses its value over a long period against the US dollar and the US dollar is consistently gaining more value against the naira and is therefore volatile, promoted non oil exports at the initial stage. The adverse effect of low exchange rates and inflation, however, discouraged export of non oil at a later stage due to hike in cost of production and other supply side constraints; poor infrastructural development, dominance of the oil sector, policy inconsistency, etc.

5.0 Conclusion and Recommendations

The purpose of this paper is to empirically investigate the impact of exchange rate volatility on export trade in Nigeria. After the literature review in the area, the paper situates itself within the premise that non oil export trade in Nigeria is predicated on a number of exogenous variables and this fact makes the fundamental approach most the suitable instrument of analysis. Time series data was collected on some key variables from 1986Q1 to 2006Q4. Unit root tests and the Johansen cointegration tests were applied.

Empirical results show evidence of stationarity at level for some variables while for some at first difference. Evidence of cointegration among the variables was also established using the Johansen procedure. This implies that a stable long run equilibrium condition exists among the fundamental variables. Error correction variable from an estimated short run dynamic model showed reasonable speed of adjustment towards the long run equilibrium path, that is, any short run disturbance, which may offset the economy along the long equilibrium path rebounds itself within two years as the evidence suggests. Furthermore, analysis of the impact of the naira exchange rate and the US dollar volatility revealed that while the former discouraged non oil exports in Nigeria, the latter promoted it by -0.885 and 1.819 for any unit change in volatility respectively. By keying this into the long run model, the naira exchange rate volatility was found to have an average adverse effect on non oil exports of -0.45 percent while the average for the US dollar volatility stood at 2.1 percent.

The paper recommends the pursuance of a sustainable and stable exchange rate policy and to put in place, measures that will promote greater exchange rate stability and improve terms of

trade conditions, promote greater openness of the economy in order to enhance non oil exports. There is the need for the government to deliver efficient infrastructural services, especially power supply and other energy resources. Lastly, while there is little that can be done to contain the effect of dollar volatility since the US continues to be one of our major trading partners, it is hoped that coming up with the above measures could greatly promote more export trade.

References

- Abrams, R. K. (1980) “Actual and Potential Trade Flows with Flexible Exchange Rates”, *Federal Reserve Bank of Kansas City Working Paper*, 80 – 01.
- Agu, C. (2002) Real Exchange Rate Distortions and External Balance Position of Nigeria: Issues and Policy Options, *Journal of African Finance and Economic Development*, Institute of African-American Affairs, New York University, New York.
- Akanji, B. (1992), “The Changes in the Structure of Export Crops and Food Crop Production under the Structural Adjustment Program: The case of Cocoa and Yam,” NISER Monograph Series.
- Aliyu, S. U. R. (1994), “Critique of Vent for Surplus Thesis and the Development of Cash Crops in Nigeria under the Structural Adjustment Programme, Unpublished M.Sc. Thesis, University of Ibadan.
- _____. (2001) ‘Trade Liberalization and Economic Growth in Nigeria: 1970 – 1998’ *Unpublished PhD. Thesis*, Faculty of Social and Management Sciences, Bayero University Kano; Nigeria.
- _____. (2007a) “Bilateral Trade Talk between Nigeria and India: A Recipe”, Being a Paper submitted for publication in the *Proceedings of a Round Table Discussion* on Nigeria-India Bilateral Talk.
- _____. (2007b) “Import-Export Demand Functions and Balance of Payments Stability in Nigeria: A Co-integration and Error Correction Modeling, Submitted to *Journal of Social and Management Sciences* (JOSAMS).
- Anderton, R. and F. Skudelny, (2001) “Exchange Rate Volatility and Euro Area Imports” European Central Bank (ECB) Working Paper, no. 64.
- Arize, Augustine C. (1998a) “The Effects of Exchange Rate Volatility on U.S. Imports: An Empirical Investigation” *International Economic Journal*, 12(3): pp. 31- 40.
- Arize, Augustine C. (1998b) “The Long run Relationship between Imports Flows and Real Exchange Rate Volatility: The Experience of Eight European Economies”, *International Review of Economics and Finance*, 7(4), pp. 417 – 435.

- Baffes, J., I. Elbadawi, and S. O'Connell. (1999) "Single-Equation Estimation of the Equilibrium Real Exchange Rate." World Bank, Policy Research Department, Washington, DC.
- Baldwin, R., F. Skudelny and D. Taglioni (2005) "Trade Effect of the Euro: Evidence from Sectoral Data, European Central Bank *Working Paper Series*, (February) No. 446.
- Baron, D. P. (1976) "Flexible Exchange Rate, Forward Markets and the Level of Trade", *American Economic Review*, 66: 253 – 66.
- Baum, C. F., M. Caglayan, and N. Ozkan (2004), "Nonlinear Effects of Exchange Rate Volatility on the Volume of Bilateral Exports", *Journal of Applied Econometrics* Vol. 19, pp. 1-23.
- Belanger, D and S. Gutierrez (1998) "Impact de la variabilite des taux de change sur le commerce international: un survol critique de la litterature", *L'Actualite economique*, 66(1): 65 – 83.
- Brada, J. C. and J. A. Mendez (1988) "Exchange Rate Risk, Exchange Rate Regime and the Volume of International Trade", *Kylos*, 41: 263 – 280.
- Bini-Smaghi, L. (1991) Exchange Rate Volatility and Trade: Why it is so Difficult to find any Empirical Relationship? *Applied Economics* 23, pp. 927-936.
- Caporale, Tony and Khosrow Doroodian (1994) "Exchange Rate Variability and the Flow of International Trade", *Economics Letters*, 46: 49 – 54.
- CBN (2007a) "Brief on the Determination of the Optimum Exchange Rate Band Under the IFEM, *External Sector Division of the Research Department*, Central Bank of Nigeria.
- CBN (2007b) "The Extent of Naira Exchange Rate Misalignment", Research and Statistics Department, Central Bank of Nigeria, October.
- CBN (2008) "The Changing Structure of the Nigerian Economy and Implications for Development", *Revised edition*, Research Department, Central bank of Nigeria, (forthcoming).
- Cerra, V. and S. Saxena. (2002) "What Caused the 1991 Currency Crisis in India?" *IMF Staff Papers*, 49 (3).
- Chang Fuzhi and Orden David (2005) "Exchange Misalignment and its Effects on Agricultural Producer Support Estimate: Empirical Evidence from India and China" International Food Policy Research Institute (IFPRI), *MTID Discussion Paper*, No. 81. Washington DC.
- Clark, P. and R. MacDonald. 1999 "Exchange Rates and Economic Fundamentals: a Methodological Comparison of BEERS and FEERS." in R. MacDonald and J. Stein (eds.) *Equilibrium Exchange Rates*, Kluwer Academic Publishers, Boston.

- Cooper, Richard N. (1971) "Currency Devaluation in Developing Countries," *Essay in International Finance*, No. 86, International Finance Section, Princeton University.
- Corrinne Ho and Robert N McCauley (2003) *Living with Flexible Exchange Rates: Issues and Recent Experience in Inflation Targeting Emerging Market Economies*, Monetary and Economic Department, BIS Working Papers, No 130
- Cote, A. (1994) "Exchange Rate Volatility and Trade", *Bank of Canada Working Paper*, 94-5
- Cottani, J., D. Cavallo and M. S. Khan (1990) "Real Exchange Rate Behavior and Economic Performance in LDCs, *Economic Development and Cultural Change*, vol.39., pp. 61-76
- De Grauwe, P. (1987) "International Trade and Economic Growth in the EMS", *European Economic Review*, 31(1/2): 389 – 98.
- De Grauwe, P. (1988) "Exchange Rate Variability and the Slowdown in Growth of International Trade." *IMF Staff Papers*.
- Dell’Ariccia, G. (1998) "Exchange Rate Fluctuations and Trade Flows: Evidence from the European Union." *IMF Working Paper*, WP/98/107.
- Diaz-Alejandro, C.F. (1984) "Exchange Rates and Terms of Trade in the Argentine Republic: 1973-76." In *Trade Stability, Technology and Equity in Latin America* eds. by IVI. Syrquin and S. Teitel, (New York: Academic Press).
- Dibooglu, S., and W. Enders, 1995, "Multiple Cointegrating Vectors and Structural Economic Models: An Application to the French Franc/U.S. Dollar Exchange Rate," *Southern Economic Journal*, Vol. 61 (April), pp. 1098–1116.
- Dixit, A. (1989) "Hysteresis, Import Penetration, and Exchange Rate Pass-Through." *Quarterly Journal of Economics*, Vol. CIV, No.2 (May) 205-227.
- Dornbush, R. and Frankel, J. (1988) "The Flexible Exchange Rate System: Experience and Alternatives" in S. Bornereditions (ed.) *International Finance and Trade*, International Economic Association and MacMillan Press, London.
- Dornbusch, R. (1987) "Exchange Rates and Prices," *American Economic Review*, Vol. 77 (March), pp. 93–106.
- Dufrenot, G and E. B. Yehoue, (2005) "Real Exchange Rate Misalignment: A Panel Cointegration and Common Factor Analysis," *IMF Working Paper*, WP/05/164.
- Edwards, Sebastian, (1986) "Are Devaluation Contractionary?" *The Review of Economics and Statistics*, (August).
- Edwards, Sebastian, (1988) *Exchange Rate Misalignment in Developing Countries*, The John Hopkins University Press, Baltimore, MD.
- Edwards, Sebastian, (1989) *Real Exchange Rates, Devaluation and Adjustment: Exchange Rate Policy in Developing Countries*, MIT Press, Cambridge, Massachusetts.

- Eichengreen, B. and D. A. Irwin (1995) "Trade Blocs, Currency Blocs and the Reorientation of Trade in the 1930s", *Journal of International Economics*, 38: 1 – 24.
- Egwaikhide, F. O. (1999) 'Determinants of Imports in Nigeria: A dynamic Specification' *African Economic Research Consortium, (AERC) Research Paper*, No. 91.
- Engle, Robert F. and C. W. J. Granger (1987). "Co-integration and Error Correction: Representation, Estimation, and Testing," *Econometrica*, 55, 251–276.
- Engle, Robert F. and K. F. Kroner (1995). "Multivariate Simultaneous Generalized ARCH," *Econometric Theory*, 11, 122-150.
- Ethier, Wilfred (1973) "International Trade and the Forward Exchange Market", *American Economic Review*, 63: 494 – 503.
- European Commission, (1995) The Impact of Exchange Rate Movements on Trade within the Single Market. *European Economy* no. 4
- Flam, H. and Jansson P. (2000) *EMU Effects on International Trade and Investment*, Stockholm University, seminar paper no.683
- Fountas, S. and K. Aristotelous (1999) "Has the European Monetary System led to more Exports? Evidence from four European Union Countries", *Economic Letters*, 62: 357 – 63.
- Franke, G. (1991) "Exchange Rate Volatility and International Trading Strategy" *Journal of International Money and Finance* 10: 292-307
- Frankel, J. and Rose, A. (2002) An Estimate of the Effect of Currency Unions on Trade and Output. *Quarterly Journal of Economics*, forthcoming.
- Frankel, J. and Wei, S. (1993) Trade Blocs and Currency Blocs. *NBER Working Paper* No. 4335.
- Gala Paulo and Claudio R. Lucinda (2006) Exchange Rate Misalignment and Growth: Old and New Econometric Evidence, *Journal of Economic Literature*.
- Ghura, D. and T. J. Greenes (1993) "The Real Exchange Rate and Macroeconomic Performance in Sub-Saharan Africa," *Journal of Development Economics*. 42: 155 – 174.
- Glick, R and A. K. Rose, (2002) "Does a Currency Union Affect Trade? The Time Series Evidence" *European Economic Review* 46(6), 1125-1151.
- Gujarati, Damodar N. (2003) *Basic Econometrics*, 4th Edition, New York: McGraw-Hill.
- Hodrick, R. J. and E. C. Prescott (1997). "Postwar U.S. Business Cycles: An Empirical Investigation," *Journal of Money, Credit, and Banking*, 29, 1–16.
- Hooper, P. & S. Kohlhagen, (1978) The Effect of Exchange Rate Uncertainty on the Prices and Volumes of International Trade. *Journal of International Trade* 8, pp 483-511

- Hirschman, Albert. O. (1949) Devaluation and the Trade Balance: A Note, *Review of Economics and Statistics*, Vol. 31, pp. 50 -53.
- Ihimodu, I. I. (1993) "The Structural Adjustment Programme and the Nigeria's Agricultural Development", *Monograph Series*, No. 2, National Centre for Economic Management and Administration (NCEMA), Ibadan, Nigeria.
- Ilimi, Atsushi (2006) Exchange Rate Misalignment: An Application of the Behavioral Equilibrium Exchange Rate (BEER) to Botswana, *International Monetary Fund*, WP/06/140
- IMF, (1984) Exchange Rate Volatility and World Trade. *IMF Occasional Paper* 28
- IMF (2005) IMF Executive Board Discusses "Fixed to Float: Operational Aspects of Moving towards Exchange Rate Flexibility", IMF Public Information Notice No.04/141.
- Johansen, Soren and Katarina Juselius (1990) "Maximum Likelihood Estimation and Inferences on Cointegration—with applications to the demand for money," *Oxford Bulletin of Economics and Statistics*, 52, 169–210.
- _____, (1992) "Testing Structural Hypotheses in a Multivariate Cointegration Analysis of the PPP and the UIP for UK," *Journal of Econometrics*, Vol. 53 (July/September), pp. 211–44.
- Johansen, Soren (1991). "Estimation and Hypothesis Testing of Cointegration Vectors in Gaussian Vector Autoregressive Models," *Econometrica*, 59, 1551–1580.
- Johansen, Soren (1995) "Likelihood-based Inference in Cointegrated Vector Autoregressive Models", Oxford: Oxford University Press.
- Kandil and Ida Mirzaie (2002) "Exchange Fluctuations and Disaggregated Economic Activity in the US: Theory and Evidence", *Journal of International Money and Finance*, No. 1 (February) pp. 1 – 31.
- Khan, M. S. (1974) 'Imports and Exports Demand in Developing Countries', *IMF Staff Working Papers*, Vol. 21, No. 3.
- Kenen, P.B. and Rodrik, D. (1986) *Measuring and Analyzing the Effects of Short-Term Volatility in Real Exchange Rates*. Review of Economics and Statistics 68(2), May 1986.
- Klaassen, F. (2004) "Why is it so Difficult to Find an Effect of Exchange Rate Risk on Trade?" *Journal of International Money and Finance*, 23, pp. 817-839.
- Koray, F. and Lastrapes, W. (1989) Real Exchange Rate Volatility and US Bilateral Trade: a VAR Approach, *The Review of Economics and Statistics* 71, pp. 708-712.
- Kroner, K. F. and W. D. Lastrapes, (1991) "Impact of Exchange Rate Volatility on International Trade: Reduced Form Estimates Using the GARCH-in-Mean Model", *Manuscript*, University of Arizona.

- Krugman, Paul, R., and Lance Taylor (1978) "Contractionary Effects of Devaluation," *Journal of International Economics*, No. 8, (August) pp. 445 – 456.
- Kwiatkowski, Denis, Peter C. B. Phillips, Peter Schmidt & Yongcheol Shin (1992) "Testing the Null Hypothesis of Stationary against the Alternative of a Unit Root," *Journal of Econometrics*, 54, 159-178.
- Lastrapes, W. & Koray, F., 1990. Exchange Rate Volatility and US Multilateral Trade Flows *Journal of Macroeconomics* 12, pp. 341-363.
- MacDonald, R., and J. Nagayasu (1998) "On the Japanese Yen-U.S. Dollar Exchange Rate: A Structural Econometric Model Based on Real Interest Differentials," *Journal of the Japanese and International Economies*, Vol. 12 (March), pp. 75–102.
- MacKinnon, James G. (1991). "Critical Values for Cointegration Tests," Chapter 13 in R. F. Engle and C. W. J. Granger (eds.), *Long-run Economic Relationships: Readings in Cointegration*, Oxford: Oxford University Press.
- MacKinnon, James G. (1996). "Numerical Distribution Functions for Unit Root and Cointegration Tests," *Journal of Applied Econometrics*, 11, 601-618.
- MacKinnon, James G., Alfred A. Haug, and Leo Michelis (1999), "Numerical Distribution Functions of Likelihood Ratio Tests For Cointegration," *Journal of Applied Econometrics*, 14, 563-577.
- Mathisen, J. (2003) "Estimation of the Equilibrium Real Exchange Rate for Malawi." *IMF Working Paper*, WP/03/104.
- Meade, James E., (1951) *The Theory of International Economic Policy, I: The Balance of Payment*, Oxford University Press, Oxford.
- McKenzie, M. (1997) "The Impact of ERV on Australian Trade Flows" *Journal of International Financial Markets, Institutions and Money*, 8:21-38.
- McKenzie, M. (1999) "The Impact of Exchange Rate Volatility on International Trade Flows", *Journal of Economic Surveys*, 13(1): 71-106.
- Meese, R., Rogoff, K. (1983) Empirical Exchange Rate Models of the Seventies: Do they Fit out of Sample? *Journal of International Economics* 14, 3–24.
- Nnanna, O. J. (2002) Monetary Policy and Exchange Rate Stability in Nigeria, Central Bank of Nigeria's *Economic and Financial Review*, Vol. 40, Number 3. pp. 1 - 22
- Obaseki, P. J. (2001) "The Purchasing Power Parity (PPP) Measure of Naira's Equilibrium Exchange Rate", *CBN Economic and Financial Review*, Vol. 36, No. 1. Pp 1 – 21.
- Ojo, O. *et al.*, (1978) "A Quarterly Econometric Model of the Nigerian Economy: Some Preliminary Estimates" *Ife Social Science Review* 1(2) 149 – 165.

- Omolola, A. (1992) "Rubber and Fisheries Industries under the Structural Adjustment Programme," *NISER Monograph Series*, No. 2.
- Omotosho, B. S. and M. U. Wambai (2005) "Is the Naira Misaligned? *Central Bank of Nigeria*
- Osagie, E. (1985), "Encouraging Production in an Abnormal Economy," in Ige & Onah (eds.), *Production Problems in the Nigerian Economy*, Uni. Jos.
- Osuntogun, A., C. C. Edordu and B. O. Oramah, (1993) "Promoting Nigeria's Non-Oil Exports: An Analysis of Some Strategic Issues", *Final Report*, African Economic Research Consortium (AERC), Nairobi, Kenya.
- Oyejide, T. A. (1986) "The Effects of Trade and Exchange Rate Policies on Agriculture in Nigeria", *Research Report 55*, International Food Policy Research Institute, Washington D.C.
- Phillips, P.C.B. and P. Perron (1988). "Testing for a Unit Root in Time Series Regression," *Biometrika*, 75, 335–346.
- Pugh, G., Tyrrell, D. and Tarnawa, L. (1999) Exchange Rate Variability, International Trade and the Single Currency Debate: a Survey. Meeusen (ed.) *Economic Policy in the European Union: Current Perspectives* (Cheltenham, UK: Edward Elgar).
- Rodrik, Dani (1994) "King Kong Meets Godzilla: The World Bank and the East Asian Miracle," in Albert Fishlow, Catherine Gwin, Stephan Haggard, Dani Rodrik and Robert Wade, eds., *Miracle or Design? Lessons from East Asian Experience*, Washington, DC, Overseas Development Council, 15 – 53.
- Rogoff, K. (1996) "The Purchasing Power Parity Puzzle", *Journal of Economic Literature* 34(2), pp. 647 - 668.
- Rose, A. (1999) "One Money, One Market: Estimating the Effect of Common Currencies on Trade". *NBER Working Paper Series*, 7432.
- Rose, A. (2000) "One Money, One Market: The Effect of Common Currencies on Trade". *Economic Policy* (30), April 2000, pages 7-33.
- Rose, A. K. and C. Engel (2002) "Currency Unions and International Integration", *Journal of Money, Credit and Banking*, forthcoming.
- Shatz, H. and D. G. Tarr (2001) "Exchange Rate Overvaluation and Trade Protection: Lessons from Experience", in Drabek, Z. (Ed), *Globalization under Threat: The Stability of Trade Policy and Multilateral Agreement*, Edgar Elgar: Cheltenham, UK.
- Sapir, A., K. Sekkat and A. Weber, (1994), "The Impact of Exchange-Rate Fluctuations on European Union Trade", *CEPR Discussion Paper*, 1041, November.
- Savvides, A. (1992) "Unanticipated Exchange Rate Variability and the Growth of International Trade", *Welwirtschaftliches Archiv*, 128: (3) 446 – 63.

Skudelny, F. (2002) Exchange Rate Uncertainty and Trade: a Survey, in: *Essays on the Economic Consequences of the European Monetary Union*, Chapter 1, Proefschrift voorgedragen tot het behalen van de graad van Doctor in de Economische Wetenschappen, Katholieke Universiteit Leuven

Soludo, C. C. and Adeola, F. Adenikinju (1997) "Exchange Rate Misalignment and Investment in Nigeria" *Paper Presented to OECD Development Centre*, Paris.

Taglioni, D. (2002) Exchange rate volatility as a barrier to trade: new methodologies and recent evidence, *Économie Internationale* 89-90 (2001-2002)

Thursby, J. G. and M. C. Thursby, (1987) "Bilateral Trade Flows, the Linder Hypothesis and Exchange Rate Risk", *Review of Economics and Statistics*, 69: 488 – 495.

Viaene, J. M. and C. G. de Vries (1992) "International Trade and Exchange Rate Volatility", *European Economic Review*, 36: 1311 – 21.

Wickens, M. (1996) "Interpreting Cointegrating Vectors and Common Stochastic Trends," *Journal of Econometrics*, Vol. 74 (October), pp. 255–71.

Williamson, J. (1994) "Estimates of FEERs," in *Estimating Equilibrium Exchange Rates*, ed. by J. Williamson (Washington: Institute for International Economics).

World Bank (1984) *Toward Sustained Development in Sub-Saharan Africa*. Washington DC: The World Bank.

World Bank, (1994), "Nigerian Structural Adjustment Programme: Policies, Implementation and Impact," May.

Yotopoulos A. Pan (1996) *Exchange Rate Parity for Trade and Development: Theory, Test and Case Studies*. Cambridge University Press.

Yotopoulos A. Pan and Sawada Yasuyuki (2005) "Exchange Rate Misalignment: A New Test of Lung-Run PPP Based on Cross-Country Data", *CIRJE Discussion Paper*, February.

Zubair, M. A. & M. I. Jega (2008) "Islamic Dinar as a Unit of Account of IDB: Implications for Competitiveness and Operational Efficiency", *Discussion Paper*, Economic Policy and Statistics Department, Islamic Development Bank, Jeddah, Saudi Arabia.

Appendix 1: Normality and Regression Residual Tests

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	9.077666	Prob. F(2,69)	0.000316
Obs*R-squared	16.87311	Prob. Chi-Square(2)	0.000217

ARCH LM Test:

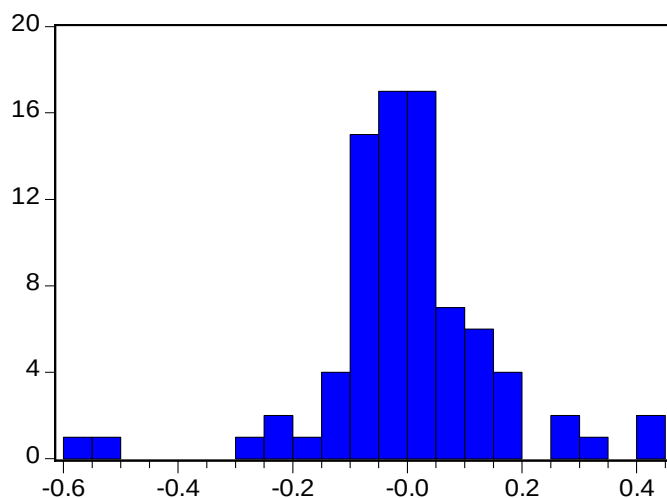
F-statistic	5.053202	Prob. F(2,76)	0.008701
Obs*R-squared	9.272318	Prob. Chi-Square(2)	0.009695

White Heteroskedasticity Test: (no Cross terms)

F-statistic	1.557841	Prob. F(18,62)	0.100785
Obs*R-squared	25.2255	Prob. Chi-Square(18)	0.11883

White Heteroskedasticity Test: (cross terms)

F-statistic	0.499819	Prob. F(54,26)	0.984056
Obs*R-squared	41.25682	Prob. Chi-Square(54)	0.898498



Series: Residuals
Sample 1986Q3 2006Q3
Observations 81

Mean 3.08e-18
Median -0.001897
Maximum 0.417663
Minimum -0.562221
Std. Dev. 0.150176
Skewness -0.454889
Kurtosis 6.645022

Jarque-Bera 47.63436
Probability 0.000000